

Integration by Parts

Evaluate each indefinite integral using integration by parts. u and dv are provided.

1) $\int x e^x dx$; $u = x$, $dv = e^x dx$

$$x e^x - e^x + C$$

2) $\int x \cos x dx$; $u = x$, $dv = \cos x dx$

$$x \sin x + \cos x + C$$

3) $\int x \cdot 2^x dx$; $u = x$, $dv = 2^x dx$

$$\frac{x \cdot 2^x}{\ln 2} - \frac{2^x}{(\ln 2)^2} + C$$

4) $\int \sqrt{x} \ln x dx$; $u = \ln x$, $dv = \sqrt{x} dx$

$$\frac{2x^{\frac{3}{2}} \ln x}{3} - \frac{4x^{\frac{3}{2}}}{9} + C$$

Evaluate each indefinite integral.

5) $\int x e^{-x} dx$

Use: $u = x$, $dv = e^{-x} dx$

$$\int x e^{-x} dx = \frac{-x - 1}{e^x} + C$$

6) $\int x^2 \cos 3x dx$

Use: $u = x^2$, $dv = \cos 3x dx$

$$\int x^2 \cos 3x dx = \frac{x^2 \sin 3x}{3} + \frac{2x \cos 3x}{9} - \frac{2 \sin 3x}{27} + C$$

7) $\int \frac{x^2}{e^{2x}} dx$

Use: $u = x^2$, $dv = \frac{1}{e^{2x}} dx$

$$\int \frac{x^2}{e^{2x}} dx = \frac{-2x^2 - 2x - 1}{4e^{2x}} + C$$

8) $\int x^2 e^{5x} dx$

Use: $u = x^2$, $dv = e^{5x} dx$

$$\int x^2 e^{5x} dx = \frac{x^2 e^{5x}}{5} - \frac{2x e^{5x}}{25} + \frac{2e^{5x}}{125} + C$$

9) $\int \ln(x+3) dx$

Use: $u = \ln(x+3)$, $dv = dx$ *or use u-sub first

$$\int \ln(x+3) dx = x \ln(x+3) - x + 3 \ln(x+3) + C$$

10) $\int \cos 2x \cdot e^{-x} dx$

Use: $u = e^{-x}$, $dv = \cos 2x dx$

$$\int \cos 2x \cdot e^{-x} dx = \frac{2 \sin 2x - \cos 2x}{5e^x} + C$$